Digital IC Design Diploma

Final System

Verification Report

RTL Simulation using ModelSim

System Overview

The system receives commands through UART RX. SYS_CTRL decodes each command and controls the Register File or the ALU. Read data and ALU results are returned through the FIFO and UART TX.

The four supported commands are Register Write (AA), Register Read (BB), ALU with operands (CC), and ALU without operands (DD).

REF_CLK is 50 MHz and UART_CLK is 3.6864 MHz. The UART uses 115200 baud with 8 data bits, even parity, and one stop bit.

The testbench starts with configuration writes, then tests register access and ALU operations. All the listed tests passed with zero errors.

Register File Tests

Configuration

Case 1 writes 81 to address 02 and 20 to address 03 before sending the normal commands. These values set the UART configuration and clock division ratio.

Sent bytes: AA 02 81, then AA 03 20.

Test Case 1 Register Write

Case 2 sends AA 04 A7 to write A7 to address 04. WrEn enables the write. The observed_reg4 signal changes from 00 to A7, confirming that the data was stored.

Test Case 2 Register Read

Case 3 sends BB 04 to read address 04. RdEn requests the read and RdData_Valid indicates valid data. RdData and the received UART reply are A7, matching the previous write.

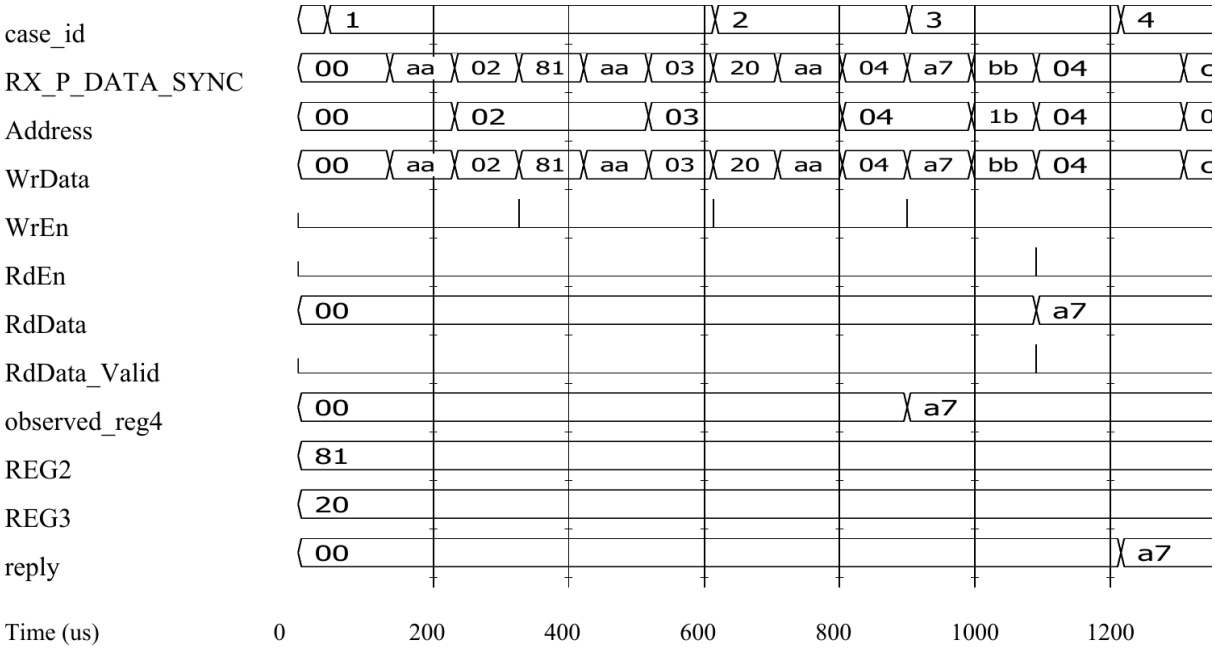


Figure 1 Configuration and Register File operations

The case_id signal identifies each stage of the testbench. Bus values in the waveforms are hexadecimal.

ALU Command Tests

Test Case 3 ALU with Operands

Case 4 sends CC 14 05 00. The command loads 20 and 5 into REG0 and REG1. ALU_FUN = 0 selects addition. The expected result is 20 + 5 = 0019.

Received bytes: 19 followed by 00. The result is correct.

Test Case 4 ALU without Operands

Case 5 sends DD 01. No new operands are sent, so the ALU uses the stored values 20 and 5. ALU_FUN = 1 selects subtraction. The expected result is 20 - 5 = 000F.

Received bytes: 0F followed by 00. The result is correct.

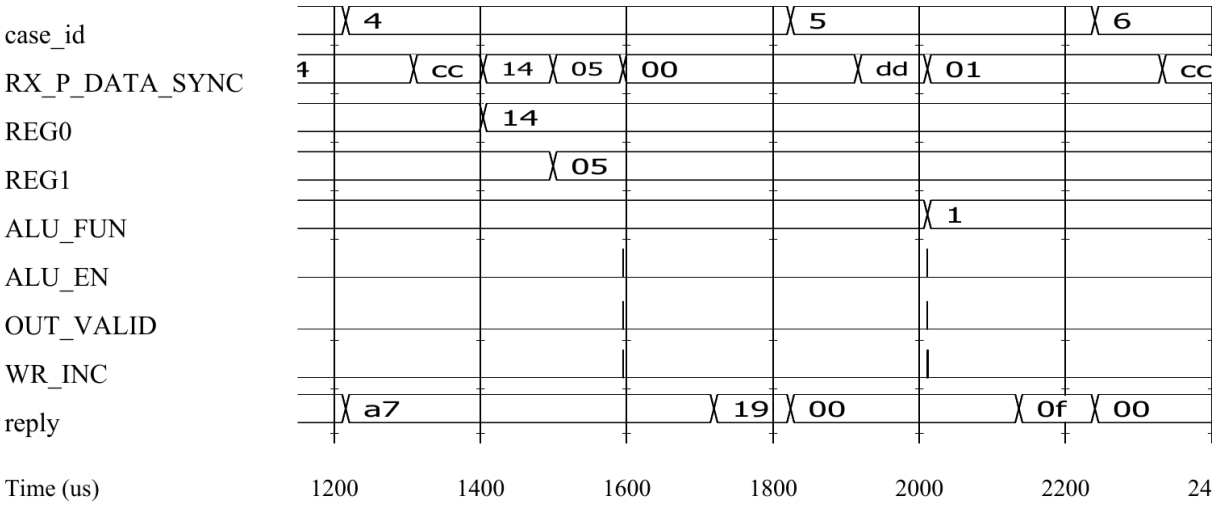


Figure 2 Addition with new operands and subtraction with stored operands

The low byte is returned first, followed by the high byte. ALU_EN starts the operation, OUT_VALID indicates that the result is ready, and WR_INC writes each result byte into the FIFO.

Additional Tests and Results

Multiplication

Case 6 sends CC FF FF 02. The result of 255 x 255 is FE01. The received bytes are 01 followed by FE, checking both result bytes.

Bitwise AND

Case 7 sends CC F3 0F 04. The result of F3 AND 0F is 0003. The received bytes are 03 followed by 00.

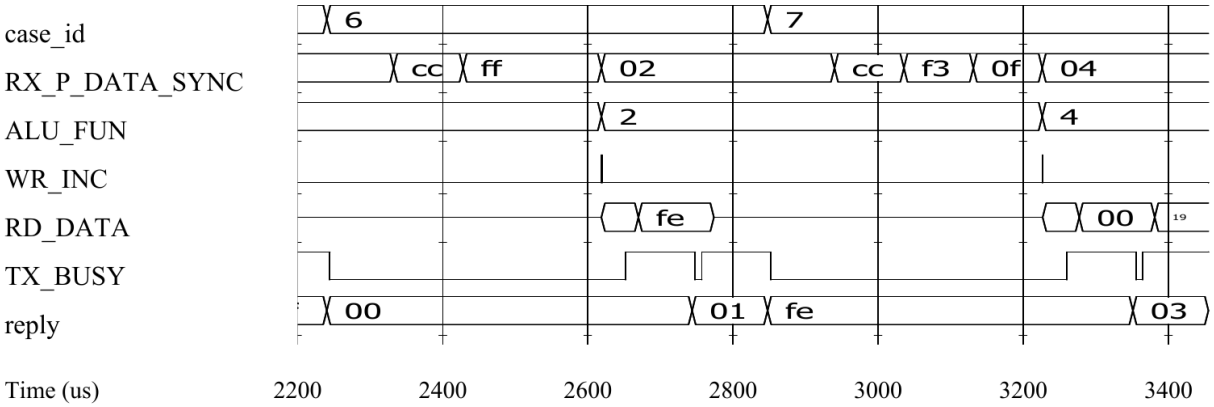


Figure 3 Multiplication and AND operations

Simulation Result

The ModelSim transcript reports the following replies:

```
PASS: case=3 reply=a7
PASS: case=4 reply=19
PASS: case=4 reply=00
PASS: case=5 reply=0f
PASS: case=5 reply=00
PASS: case=6 reply=01
PASS: case=6 reply=fe
PASS: case=7 reply=03
PASS: case=7 reply=00
SUMMARY: errors=0
ALL TESTS PASSED
```

The testbench completed the sequence and stopped at 3.455547208 ms. TX_BUSY shows UART transmission, while reply holds each received byte.